

Patricio Antonio García Pérez Vela

Software Engineer | Junior Python Developer & Full-Stack Web

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Profile

Graduate Computer Systems Engineer specializing in full-stack Python web development (FastAPI, Flask, Django) and TypeScript/Next.js. Proficient in SQL relational database design and maintenance (PostgreSQL, SQL Server), RESTful API engineering, Docker containerization, cloud deployment (GCP), and CI/CD pipelines (GitHub Actions). Proven experience gathering user requirements, drafting user stories, estimating tasks, and creating technical documentation under Agile (Scrum) methodologies. Advanced English (B2).

Technical Skills

Python & Web Development: Python (OOP, Data Structures, Control Flow), FastAPI, Flask, Django, REST APIs, TypeScript, JavaScript, React, Next.js, HTML5, CSS3.

Databases (SQL): PostgreSQL, SQL Server, Supabase, SQLite, SQLAlchemy, Entity Framework, Alembic, Relational Schema Design.

Cloud, DevOps & Tools: Google Cloud Platform (GCP), Docker, CI/CD (GitHub Actions), Git, GitHub, Vercel, Render, Linux, WSL2.

Software Engineering & Agile: Scrum, Agile SDLC, Requirements Gathering, User Stories, Estimations, Technical Manuals.

Languages: Native Spanish, Advanced English (B2 – Full Professional Proficiency).

Professional Experience

Mazda Motor Manufacturing de México

Software Engineer Intern – IT Document Management Digitalization FY25

Salamanca, Gto., Mexico

Aug 2025 – Feb 2026

- Digitalized critical IT processes previously relying on Excel/SharePoint into a centralized SQL Server & C#/.NET platform with full auditable traceability and duplicate prevention.
- Led user requirement gathering with stakeholders, authored user stories, calculated delivery estimations, and produced technical manuals under Agile practices.
- Modeled the relational database schema and Entity Framework data access layer, enforcing Role-Based Access Control (RBAC) integrated with Active Directory.
- Streamlined metadata update duration from 20 to 3 minutes per request and automated traceability log generation.

Stack: Python, C#/.NET, SQL Server, Entity Framework, Active Directory, User Stories, Scrum, Technical Documentation

Selected Projects

Electro-Scorpio Control – Electrification Contract & Material Matrix Management

2026

GitHub (p5Patricio/electro-scorpio-control-back) | Full-Stack Python

- Architected and developed a full-stack web application with a Python (FastAPI) backend and PostgreSQL (SQLModel) database for contract management and material allocation matrices.
- Engineered the responsive frontend in TypeScript with Next.js, consuming REST endpoints with strict Pydantic v2 schemas and JWT security.
- Designed an 8-table relational SQL database to record daily field progress logs, financial estimations, and inventory tracking.

Stack: Python, FastAPI, SQLAlchemy, PostgreSQL, TypeScript, Next.js, HTML/CSS, REST APIs, Git/GitHub

DEMOX – Political Intelligence Platform

2026

GitHub (p5Patricio/Demox-Backend) | Full-Stack + Cloud

- Engineered backend web services in Python (FastAPI) backed by PostgreSQL/Supabase and asynchronous background processing via Celery and Redis.
- Containerized applications using Docker and configured CI/CD pipelines via GitHub Actions for automated testing and cloud deployments (GCP/Render).

Stack: Python, FastAPI, Next.js (TypeScript), PostgreSQL, Docker, CI/CD (GitHub Actions), GCP, Celery, Redis

WhisperKey – Offline Voice Dictation App in Python

2026

github.com/p5Patricio/WhisperKey

- Developed a 100% local voice dictation application utilizing OOP principles in Python, multi-threaded GPU inference, and cross-platform executable packaging.

Stack: Python (OOP), Whisper, CustomTkinter, CUDA, PyInstaller, Git/GitHub

Education

Universidad de Guanajuato (DICIS)

Bachelor of Science in Computer Systems Engineering – Graduated

Database Systems, Full-Stack Web Development, and Algorithms.

Salamanca, Gto., Mexico

Dec 2025 GPA: **9.4/10.0**. Focus: Software Architecture, SQL